# Supplementary Information: 700-km intercity Yb/Sr optical clock comparison at $2.3\times10^{-18}$

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## 1 Per-segment data

Table 1: Per-segment data of the 16 segments analyzed in this work. Ratio deviations $y_i = R_i/R_1 - 1$ are relative to the first segment, with $R_1 = 1.207\,507\,039\,343\,337\,721\,0$. $u(y_i)$ is the statistical uncertainty of the segment ratio; "Fit" is *est.* for segments whose stability fit lies in the white-frequency-noise band and *lim.* for model-limited fits.

| Segment | $n_{\mathrm{valid}}$ | $y_i$ ($10^{-18}$) | $u(y_i)$ ($10^{-18}$) | Fit |
|---|---|---|---|---|
| 1 | 68 009 | +0.00 | 4.245 | est. |
| 2 | 29 481 | -3.36 | 3.017 | lim. |
| 3 | 42 891 | +2.00 | 5.880 | est. |
| 4 | 64 280 | -4.61 | 3.674 | est. |
| 5 | 19 532 | -3.38 | 15.250 | est. |
| 6 | 52 140 | -1.15 | 2.230 | lim. |
| 7 | 39 526 | +5.31 | 3.476 | lim. |
| 8 | 43 565 | +0.97 | 9.860 | est. |
| 9 | 57 938 | -0.76 | 5.003 | est. |
| 10 | 35 824 | +2.61 | 5.976 | est. |
| 11 | 40 346 | +0.73 | 4.843 | lim. |
| 12 | 152 994 | -3.09 | 4.019 | est. |
| 13 | 68 618 | +1.97 | 6.254 | est. |
| 14 | 57 649 | -0.25 | 4.028 | est. |
| 15 | 147 640 | -0.77 | 1.809 | lim. |
| 16 | 65 983 | -0.89 | 8.813 | est. |

## 2 Within-segment tidal correlation analysis

## 3 Frequency-chain relations and the ratio inversion

The two clock transitions are bridged to the 1 550 nm telecom carrier through fixed factor-of-two relations. On the Shanghai side the Sr 698 nm transition is doubled to 1 397 nm; on the Wuhan side the Yb 578 nm transition is doubled to 1 156 nm. Both ends are phase-locked to independent optical frequency combs whose repetition rate is $f_{\mathrm{rep}} = 200$ MHz, derived from a 10 MHz reference divided by 20.

The recorded observable is the out-of-loop beat $b$ between the received carrier and the local Yb-referenced comb, sampled every second. The ratio inversion uses the fixed chain constants

Table 2: Leave-one-out sensitivity of the ratio combination. Each row excludes one of the 16 segments and recomputes the combination from the remaining segments. The center values are stable at the $10^{-18}$ level under any single exclusion.

| Excluded segment | $R_{\mathrm{wls}}$ | $\chi^2_{\mathrm{red}}$ | $u_{\mathrm{wls}}$ $(10^{-18})$ |
|---|---|---|---|
| 1 | 1.207 507 039 343 337 720 4 | 0.542 | 1.136 |
| 2 | 1.207 507 039 343 337 720 8 | 0.474 | 1.164 |
| 3 | 1.207 507 039 343 337 720 3 | 0.531 | 1.123 |
| 4 | 1.207 507 039 343 337 720 8 | 0.451 | 1.145 |
| 5 | 1.207 507 039 343 337 720 4 | 0.540 | 1.111 |
| 6 | 1.207 507 039 343 337 720 6 | 0.535 | 1.217 |
| 7 | 1.207 507 039 343 337 719 9 | 0.347 | 1.150 |
| 8 | 1.207 507 039 343 337 720 4 | 0.542 | 1.114 |
| 9 | 1.207 507 039 343 337 720 4 | 0.543 | 1.128 |
| 10 | 1.207 507 039 343 337 720 3 | 0.525 | 1.122 |
| 11 | 1.207 507 039 343 337 720 4 | 0.539 | 1.130 |
| 12 | 1.207 507 039 343 337 720 6 | 0.512 | 1.139 |
| 13 | 1.207 507 039 343 337 720 4 | 0.533 | 1.121 |
| 14 | 1.207 507 039 343 337 720 4 | 0.543 | 1.139 |
| 15 | 1.207 507 039 343 337 720 6 | 0.540 | 1.287 |
| 16 | 1.207 507 039 343 337 720 4 | 0.543 | 1.115 |

| Constant | Value | Meaning |
|---|---|---|
| $N_{1156}$ | 1,295,739 | Yb 1156 nm comb count |
| $N_{1397}$ | 858,456 | Sr 1397 nm comb count |
| $N_{1550}$ | 773,598 | 1550 nm link count (Shanghai end) |
| $N_{1550}^{\mathrm{WH}}$ | 966,996 | 1550 nm link count (Wuhan end) |
| $f_{\mathrm{ref}}$ | 10 MHz | comb repetition base |
| $\mathrm{div}_{20}$ | 20 | divider ($f_{\mathrm{rep}} = 200$ MHz) |

The beat is converted to fractional units with the transfer frequency $f_{1550} = 193.3992\,\mathrm{THz}$ $(= N_{1550}^{\mathrm{WH}}\, f_{\mathrm{rep}})$. The segment-mean beat deviation $\overline{\delta b}$ enters the Sr/Yb ratio through

$$\mathrm{Dr} = \frac{c_{1156}}{N_{1156}} \frac{\overline{\delta b}}{f_{\mathrm{ref}}\,\mathrm{div}_{20}} \Bigg/ \left[\frac{c_{1397}}{N_{1397}}\left(N_{1550} + \frac{7}{25} + \frac{1}{25}\right)\right],$$

where $c_{1156} = (1 + b_{\mathrm{Yb}})/2$ with $b_{\mathrm{Yb}} = -1.7 \times 10^{-18}$ (Yb-side fixed shift) and $c_{1397} = (1 + a)/2$, with $a$ collecting the five Sr systematic components listed below. The segment ratio is

$$\mathrm{Sr/Yb} = \mathrm{ratio}_{\mathrm{base}} + \mathrm{Dr}, \qquad \mathrm{ratio}_{\mathrm{base}} = \frac{c_{1156}}{N_{1156}} \frac{N_{1550}^{\mathrm{WH}} + \frac{26}{20} + \frac{m}{f_{\mathrm{ref}}\,\mathrm{div}_{20}}}{\frac{c_{1397}}{N_{1397}}\left(N_{1550} + \frac{8}{25}\right)},$$

where $m$ is the campaign-global median of the plausible beat samples. The static gravitational-redshift correction is applied as a multiplicative factor,

$$(\mathrm{Yb/Sr})_{\mathrm{corr}} = (\mathrm{Yb/Sr})_{\mathrm{raw}}\,(1 + \delta_g), \qquad \delta_g = -3.116 \times 10^{-15}.$$

All ratio arithmetic is carried out in 80-digit decimal precision, mirroring the campaign processing convention.

The Sr systematic shift $a$ is the sum of five components evaluated per segment: lattice AC-Stark ($a_{\mathrm{rou}}$, about $-2.3$ to $-3.5 \times 10^{-19}$), AC Stark ($a_{\mathrm{AC}}$, $+7.44$ or $+8.93 \times 10^{-18}$ by campaign period), second-order Zeeman ($a_{\mathrm{SM}}$, the dominant term), air refractive index ($a_{\mathrm{air}} =$

$-6.7 \times 10^{-19}$), and blackbody radiation ($a_{\mathrm{BBR}} = 0$). The first three are segment-dependent constants recorded in the campaign parameter file; the last two are fixed. Full per-segment values are reproduced in the frozen result layer referenced in Data availability.

## 4 Statistical model and combination

Each segment $i$ contributes a ratio estimate $R_i$ and a statistical uncertainty $u_i$ obtained from the overlapping Allan deviation $\sigma_y(\tau)$ of the beat fractional frequency, evaluated on a tau grid of powers of two plus 600, 1200, 3600 and 7200 s, limited to a quarter of the longest gap-free run of the segment. $u_i$ adopts the last computable point, $u_i = R_i\,\sigma_y(\tau_{\max})$, without extrapolating the stability curve beyond the data; the log-log slope of $\sigma_y(\tau)$ is retained as a white-frequency-noise diagnostic only.

The segments are combined under a random-effect model with common excess scatter, $R_i \sim \mathcal{N}(\mu, u_i^2 + \xi^2)$. Three estimators are evaluated:

- *Weighted least squares* with inverse-variance weights $w_i = 1/u_i^2$: $\hat{\mu} = \sum w_i R_i / \sum w_i$, with $u_{\mathrm{wls}} = 1/\sqrt{\sum w_i}$. The reduced chi-square $\chi^2_{\mathrm{red}} = \sum (R_i - \hat{\mu})^2/u_i^2/(N-1)$ and the Birge ratio $\sqrt{\chi^2_{\mathrm{red}}}$ quantify the segment agreement.
- *Mandel–Paule*: the same observation model with $\xi$ chosen so that $\chi^2_{\mathrm{red}}(\xi) = 1$.
- *Bayesian random effect*: flat priors on $\mu$ and on $\log\xi$, with the joint posterior marginalized numerically over a 40-decade $\log\xi$ grid (401 points, trapezoidal quadrature). The grid span is expanded until the posterior mass at the edges falls below 1%. The posterior mean of $\mu$ and its posterior standard deviation give the reported central value and statistical uncertainty.

All three center values are reconstructed in decimal precision from the segment-1 baseline, $R = R_1\,(1 + \hat{\mu}/R_1)$, so that differences at the $10^{-18}$ level are not lost in floating-point arithmetic.

## 5 Leveling details and national-standard comparison

The first-order spirit leveling line runs about 1 113 km between the Wuhan benchmark WS-03 and the Shanghai benchmark WS-01 through Ezhou, Huangshi, Huanggang, Lu'an, Hefei, Wuhu, Guangde and Suzhou. Digital levels with bar-code invar staffs were used, with sight lengths not exceeding 30 m; the line was observed in 6 km to 8 km sections in both directions, each checked in the field, and gravity was observed along the line at the leveling points with two relative gravimeters. The per-kilometer accidental error is $M_\Delta = 0.255\,\mathrm{mm\,km^{-1}}$. The most stringent first-order requirements of the national standards of China (GB/T 12897–2006, $M_\Delta \le 0.45\,\mathrm{mm\,km^{-1}}$), the United States (FGCC 1984, $0.5\,\mathrm{mm\,km^{-1/2}}$), Germany (AdV FA-PN 2022, 0.4 mm) and Czechia (ČÚZK 2015, $m_0 = 0.40\,\mathrm{mm\,km^{-1}}$) are all satisfied.

The independent GNSS-plus-geoid check used precise point positioning for the ellipsoidal heights of the permanent marks, local leveling for the vertical offsets to the clock reference surfaces, and the EGM2008 geoid model for the conversion to geopotential, giving $280.60 \pm 2.558\ \mathrm{m^2\,s^{-2}}$, consistent with the leveled value but about 30 times less precise, so it serves as a consistency check rather than an independent calibration at the reported accuracy.

## 6 Tidal template construction and the two tide models

The primary tide template is the solid Earth tide and ocean tide loading displacement difference between the two sites, computed every 30 s with the public software ETideLoad4.5, which implements the geophysical models and numerical standards of the IERS Conventions (2010). The

displacement difference is converted to an equivalent geopotential difference with the local gravity $g = 9.794\,\mathrm{m\,s^{-2}}$ and then to a beat-frequency template $h = f_{1550}\,\Delta W/c^2$ with the same $f_{1550}$ normalization as the beat record; the direction convention is $\Delta W = W(\mathrm{Wuhan}) - W(\mathrm{Shanghai})$.

The cross-check template computes the solid tide from the DE421 ephemerides (Earth–Sun–Moon positions) with Love numbers, and takes the ocean-tide loading correction to the gravitational potential directly from the FES2022 global ocean-tide model, on a 90 s Beijing-time grid. The two templates are independent realizations; their mutual correlation over the campaign is +0.88, and their segment-mean predictions differ in detail (Figure 1b).

## 7 Transfer subsystems

The fiber route is divided into spans of tens of kilometers, each terminated by a relay station carrying a narrow-linewidth laser, an erbium-doped fiber amplifier and a fiber-noise-cancellation loop; 21 independent relay stations carry 22 amplifiers in total. The link extends the field-deployed cascade of a previous 2 067 km demonstration, and its added instability is kept below the clock noise for averaging times beyond 100 s. Both frequency combs are self-referenced and phase-locked to their local clock signal; all other frequencies in the chain derive from the two clock transitions through the fixed factor-of-two relations of Section 3, so the two clocks are the system's only independent frequency sources. The fiber loop-back is monitored out of loop throughout. The frequency-conversion stages each contribute $0.1\times10^{-18}$ and $0.1\times10^{-18}$ to the uncertainty budget from their frequency-reference settings.

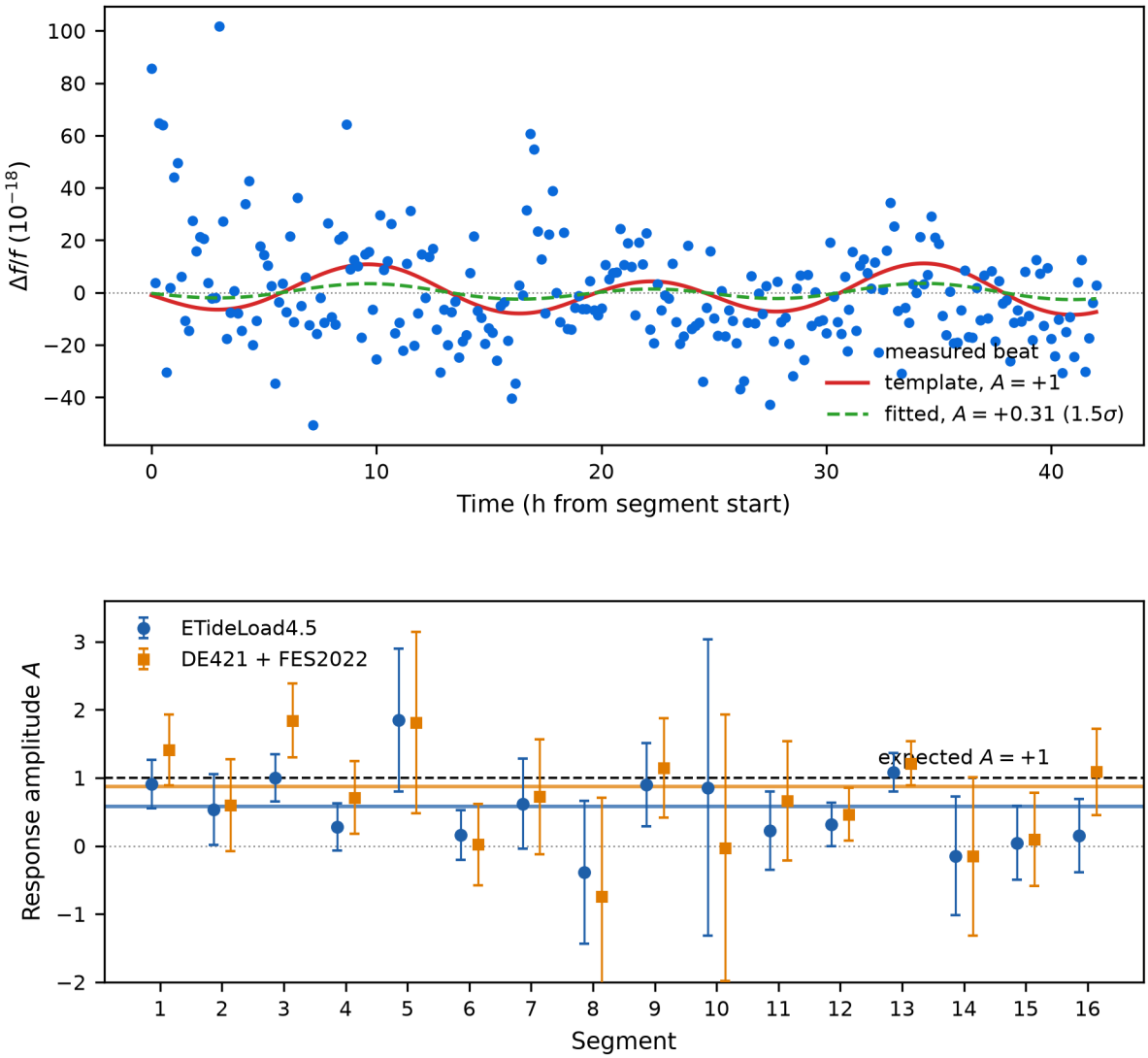



Figure 1: Within-segment tidal correlation analysis. **(a)** Single-segment example: the 1200-s demeaned beat waveform (dots) and the tidal template scaled at the expected response $A = +1$ and at the fitted amplitude, for the ETideLoad4.5 model. No single segment reaches a significant correlation on its own. **(b)** Per-segment response amplitudes $A$ for the two tide models, over the 16 segments entering the ratio combination. Error bars are per-segment fit uncertainties from non-overlapping windows; the solid lines mark the precision-weighted combinations. In 12 of the 16 segments the correlation is positive for both models, and both precision-weighted amplitudes (+0.58 and +0.87) fall short of the expected response of +1 (dashed line), so the observed tidal coupling is smaller than any single template predicts.